

Concurrency Basics

AI Tutor - Golang - Lesson 3

Goroutines

A goroutine is a lightweight thread managed by the Go runtime. Starting one is as simple as adding the 'go' keyword before a function call, letting it run concurrently with the rest of the program.

- `go doSomething()`

Channels

Channels let goroutines communicate safely by sending and receiving values, avoiding the need for manual locks in many cases:

- `ch := make(chan int)`
- `go func() { ch <- 42 }()`
- `value := <-ch`

Why It Matters

Concurrency lets a Go backend handle many requests at once efficiently - for example, a web server can process hundreds of API calls in parallel using goroutines without blocking.